Calculus of Variation Notes

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Definition 1 (Functional) a functional J maps a function h to a real number
Definition 2 (Continuity of functional) The functional $J[y]$ is continuous at point $\hat{y} \in \mathcal{C}$(normed linear space) if for any $\epsilon > 0, \exists \delta > 0$ such that

$$|J[y] - J[\hat{y}]| < \epsilon, \quad \text{provided that } ||y - \hat{y}|| < \delta$$

The set of functions satisfying the constraints of a given variational problem are called the **admissible functions/curves**.

1. Given a functional, we should choose a function space such that the functional is continuous
2. Admissible functions in general is not a linear space. (Sum of two functions violates the constraints)

Linear functional ϕ on $\mathcal{C}(a, b)$ examples:

1. $\phi[h] = h(x_0)$ for a fixed point $x_0 \in [a, b]$
2. $\phi[h] = \int_a^b \sum_i \alpha_i(x) \frac{d^i h(x)}{dx^i} dx$.

Definition 3 (Differential/Variation of a functional) Let J be defined on a normed linear space, let

$$\Delta J[h] = J[y + h] - J[y]$$

be its increment. If y is fixed, then $\Delta J[h]$ is a functional of h. Suppose

$$\Delta[h] = \phi[h] + \epsilon||h||$$

where $\phi[h]$ is a linear functional.